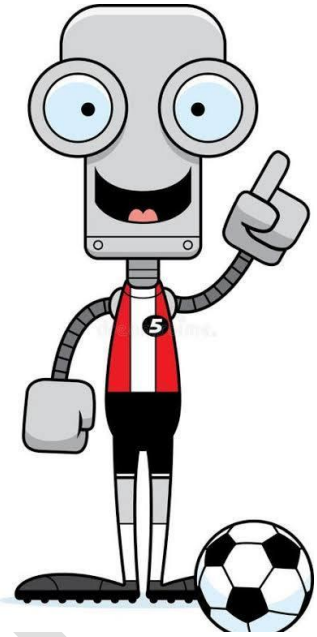
Project – Robot Soccer



Learning Objectives:
The purpose of this project is to provide an opportunity to apply all that you have learned in an open-ended design problem. The task is to design and build a machine that can compete in a robotic interpretation of soccer.

Specifications:
Your robot must be a stand-alone entity, capable of meeting all specifications described in this document. Battery power will be required. You are limited to a total team expenditure of $200 for the robot, beyond the parts in your kit.

- Your robot will navigate the Robot Soccer field and attempt to move the ball into the opposing goal. Each match will last 4 minutes.

- The ball will be ~7 inches in diameter, inflatable red dodge ball. It has a mass of 275 grams.

- At the beginning of each match, or upon a field reset, the ball will be placed in the center of the field, equidistant from the two goals.

- The playing field will be a rectangular, 12' x 8' puzzle-piece foam exercise mat. The mat is ½" thick black (actually very dark gray) modular foam flooring, textured side up. The diagram is not to scale.

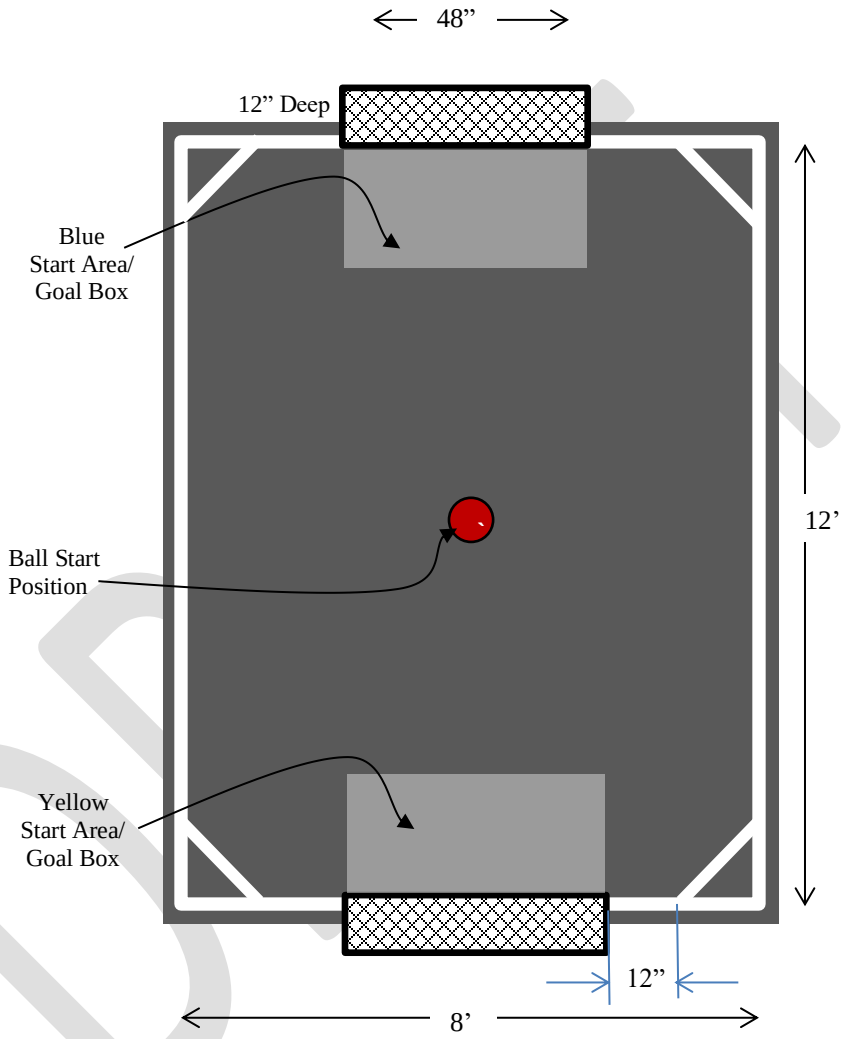
- The goals will be 24" high and 48" wide, and 12" deep with a sloped bottom, to attempt to 'suck in' balls in that cross the goal line. You will defend your goal, and attempt to score in the opposing goal. Colored cards will be placed on the top corners of the goals to allow robots to identify and locate them.

- You will be designated as either the 'BLUE' or 'YELLOW' robot and your opponent will be the opposite color. A toggle switch can be included with your robot that will switch your robot from YELLOW mode to BLUE mode. Your robot will also 'wear' a knit cap colored blue or yellow, corresponding to your team color.

- The referee will determine when the ball has been scored in the goal (or we will automate detection if possible). After each scored goal, both robots and the ball will be returned to their starting positions (field reset) and the match will continue. If the 4-minute match ends in a tie, an additional 90 seconds of overtime will be played. If the match is tied after overtime, a penalty shootout will be implemented. The teams will alternate with three shots each for the initial shoot out. If the score is still tied, the robots will alternated shots until one wins.

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- In front of each goal will be a goal box painted a lighter gray than the main field. The goal box will be the same width as the goal (48") and extend 24" from the goal.



- To start, your robot will be placed in your color's white goal box at a random orientation. A referee will activate your starting switch to begin the contest. We will place the robots to one side of the goal box, so that your robot should be able to see the ball, your goal, and possibly the opposing goal. Once the robot has been placed and the switch activated, no member of the team may touch the robot until the contest is over, or the field is reset. The goal box is not actually painted your color; both goal boxes are light gray as noted above.

- The perimeter of the playing field will be marked off with 2" wide vinyl tape, and an additional 2" of bare black mat will be left beyond the white tape to allow the ball some room to strike the perimeter barrier.

- A perimeter barrier will enclose the mat. It will be constructed from thick-walled PVC tube and elevated 5 inches above the playing surface. If the ball travels over the perimeter barrier, it will be replaced in the center of the field for a field reset. The barrier will be angled a foot from each corner.

- You may include a field reset switch on your robot that you can press when the field is reset after a goal, for a penalty kick, new match, etc. You can have multiple switches to indicate the condition, if you like.

- Your robot should fit into a 14" x 12" rectangular footprint and should not be any taller than 12" at its highest point. A vertical, rigid perimeter must enclose the robot from 6" above the floor to 8" above the floor. This is the 'bumper,' intended to ensure that all robots will contact each other on the bumper, and not elsewhere. All other parts of the robot (wheels, kicker, camera, etc.) must lie inside the bumper when viewed top down. Any extension beyond the perimeter must return within the perimeter within 2 seconds of when it initially extended.

- A 2-inch high ring must be installed from the 10 inches off the ground to 12 inches. The ring should be circular and 8 inches in diameter. It can also be a polygon of at least 8 sides, 8 inches across. Your cap (blur or yellow) will rest on this ring.

- Parts of the robot may extend beyond the bumper envelope during operation as long as they remain connected to the robot, return within the envelope during normal operation, and do not obstruct the vision or mobility of other robots.

- One or more 'kicker' mechanisms may be employed on your robot. They each require a failsafe to ensure your robot does not 'kick' an opponent. A robot that does kick an opponent grants the opposition a penalty kick.

- Your robot, including extensions, cannot grab the ball with grippers or pincers, or otherwise trap, envelop or capture the ball. This precludes any device that surrounds or captures more than half the ball, as seen from any direction.

- Your robot cannot lift the ball off the ground, but can 'kick' it or propel it so that it might leave the ground as a result.

- If the robots are in direct contact and attempting to move through each other ('locked up') the referees can call for a field reset.

- Your robot cannot take any action perceived by the referee to cause intentional damage to the playing area or an opposing robot. The referees can award penalty

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shots to your opponent if your robot appears to be intentionally damaging the opposing robot.

- During a penalty shot chance, the penalized robot shall be removed from the field, the ball placed at its center, starting position, and the penalty taking robot shall be placed at the ***center*** of its goal box (as opposed to the side of the goal box for a generic field reset). The penalty taking robot shall have 15 seconds to attempt to score on the open net. If it scores, the goal is awarded and the field is reset. If it does not score in time, both robots are replaced and the field is reset. The time for a penalty kick does not count toward the 4 minute match length.

- Intentional 'flopping' shall also result in a penalty shot. Robots that tip over at the slightest nudge, engineers that intentionally design the bumper to shatter easily, or other actions or Neymar-like design strategies that attempt to make the opposing robot appear dangerously aggressive will be penalized for flopping at the referees' discretion.

- Your team members and the referees will need to pick up your robot. Please include a handle or handles to allow your robot to be easily picked up without causing damage.

- Compressed air or carbon dioxide may be used to create pneumatic actuators. The pressure must be regulated to 85psi (or lower) for the pneumatic system, but may have higher pressure commercial vessels on board. You must have 'secondary containment,' i.e., an independent enclosure, around any pressure reservoirs to prevent damage to the vessel. Appropriately rated PVC pipes work well for this.

- Your robot cannot damage the floor, ball or the walls in any way: no studded tires, digging into the floor, use of adhesives (on wheels or ball), emissions (infrared, ultrasonic or otherwise), or extension of obstructing parts, ballistic parts, electro-magnetic pulse, shocks, flames, chemicals, etc. Use of any of these is grounds for disqualification.

- Your robot must stay in one piece and any extension beyond the bumper envelope must retract back inside the envelope as part of its movement process.

- Your robot may not alter the playing field in any way.

- You may have a maximum of three microprocessors, microcontrollers or similar devices on board. A Teensy, Arduino, Beagle Bone, mBed, Netburner, Altera DE2, RPi, Jetson, or NI sbRio board would each count as one micro. Simple, small, cheap micros can be counted as ICs if they have limited functionality, e.g., a battery-monitoring controller.

- You may have up to two camera systems on board. A Pixy, Luxonis, or Pi Camera each count as one. IR sensor pairs do not count.

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- A report describing the technical details of your robot will be required. The report should be of sufficient detail that a person skilled at the level of a Mechatronics graduate could understand, reproduce and modify your design.

- Your robot must be safe for the user, the spectators, and other robots. When your robot interacts with the ball, it should not travel higher than 2' vertically at any time. Please be VERY, VERY careful and **conscientious** when testing your actuators.

Evaluation:
There will be six evaluation periods or milestones in this project.

1. **Friday, September 18th. Prototypes and design plans due.**
   You must present an informal design plan (no more than 6 pieces of 8.5 x 11 paper, no less than 2). This should detail the plans for the drive system, sensor system (camera, IR, sonar, radio, etc.) and ball interaction mechanism. Specify materials and manufacturing methods. You are **not** obligated to carry out these designs exactly. Prototypes are highly encouraged.

2. **Monday, October 19th. Demonstration of independent system functionality.**
   You must have all subsystems running independently, with appropriate micro-controller input or output. The subsystems do not have to function together, but should work correctly and reliably with portable power sources. Radio system, ball mechanism, drive system, vision system (if applicable), etc., all must function correctly.

3. **Monday, November 2nd. Complete functionality.**
   Your robot should be fully functional. You should be able to navigate the field and interact with the ball and goals. Your code may be simple or rudimentary, but it should work.

4. **Monday, November 16th. Dry run.**
   This is a practice run for your robot. You should be able to maneuver and score goals at this point.

5. **Wednesday, December 2nd. Project runoff.**
   The runoff begins at 4pm. This is the main graded portion of the project. Robots will be randomly seeded for matches. Allowances may be made for malfunctions in early matches.

6. **Friday, December 4th. Project report.**
   The report should be of sufficient detail that a person skilled at the level of a Mechatronics graduate could understand, reproduce and modify your design. Schematics, mechanical drawings, labeled digital photos and well-commented code listings should accompany your descriptions of each subsystem that makes up your robot. Clear, complete and concise are your goals. Include a section on your design process, and a short summary of lessons-learned, which students next

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year might find valuable. As a rule of thumb, the body of the report should be around 10 pages, with mostly labeled diagrams and text to support them, and a maximum of 10 pages of appendices.

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Grading:
There will be four team grades for this project.

1. **Concept (10%)**
   This will be based on the technical merit of the design and coding for the machine. Your machine will be evaluated on the appropriateness of the solution, innovative hardware, software and application of physical principles, in the solution.

2. **Implementation (10%)**
   This will be based on the machine as presented at the runoff. Your robot will be evaluated on physical appearance and quality of construction. Aesthetics, craftsmanship and finished appearance will be the primary criteria.

3. **Report (20%)**
   The report should be of sufficient detail that a person skilled at the level of a Mechatronics graduate could understand, reproduce and modify your design. Schematics, mechanical drawings, labeled digital photos and well-commented code listings should accompany your descriptions of each subsystem that makes up your robot. Clear, complete and concise are your goals.

4. **Performance evaluation (60%)**
   This will be the combined evaluation of your robot at the various milestones and evaluation periods. You will be graded on whether your robot meets the requirements at the independent system function demonstration, complete functionality demonstration, dry run and runoff milestones. At least 20% of the evaluation will be based on your robot's performance at the runoff.

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